CAPACITOR CIRCUIT — BLINK WITHOUT A CHIP!

THE FADING TORCH MYSTERY

The torch was switched off… but it kept glowing for a moment!
Where does the extra light come from?
Something inside was storing energy — and slowly letting it go.
🔊 Teacher: "Zzzzt… Flash!"

"Where does the extra light come from?"

MEET THE STORAGE TANK!

Capacitor = Temporary Electricity Water Tank

Just like a Sintex rooftop tank stores water for a short time, a capacitor stores electricity! It fills up quickly and pours it out fast — making the LED blink!

Like rooftop water tanks store water for a short time — capacitors store electricity briefly!

BATTERY VS CAPACITOR

Battery fills slow, capacitor fills & empties fast! Both work together to power your circuit!

🔋 Battery = Deep well (Main Power Source)

⚡ Capacitor = Small cup (Storage Tank)

LET'S PRETEND!

- Fill the cup! Slurrrp...
- Pour the water! Splash!
- Now YOU are the capacitor!
- Fill up slow... then POUR it all at once!
- Teacher cue: "Slurrrp... Splash... Blink!"

Fast fill, fast pour = LED blinks! ⚡

CHARGING & DISCHARGING EXPLAINED!

A small bucket fills slowly under a leaking tap — then tips over all at once! That is exactly how a capacitor works with electricity.

Charging = filling the bucket from the tap (slowly storing energy).

Discharging = bucket pours water fast — LED blinks!

THE TINY FESTIVAL LIGHT

Meet the LED!

LED = Tiny festival light that glows with little energy. Just like mirchi lights on a rice string, LEDs are small, bright, and use very little power!

Spot the LED glowing in our blink circuit!

CIRCUIT PARTS — MEET THE TEAM!

Every circuit has a team of parts working together! Let's meet each one and find out what job it does to make the LED blink!

🔋 9V Battery
= Main power source

⚡ Capacitor
= Storage tank

🟫 Resistor
= Energy traffic cop

💡 LED
= Tiny festival light

BUILD THE BLINK CIRCUIT!

Follow each step carefully! Connect the parts in order and watch your LED blink — no chip needed!

⚡ Step 1: Connect the 9V battery snap clip to the resistor leg.

🔧 Step 2: Connect resistor to the capacitor cylinder top.

🔌 Step 3: Connect the other resistor leg to the capacitor (+) pin.

💡 Step 4: Connect capacitor to LED — watch it blink as it fills & pours!

FIX THE BLINK! QUIZ

Something is wrong with the blink circuit! Can you guess what happened? Look at each clue and think carefully!

Bigger tank (larger capacitor) → LED stays on longer. Why?

No battery → No blinking, LED fades immediately. What is missing?

SAFETY FIRST!

Before we build our blink circuit, let's make sure we stay safe and have fun! Follow these simple rules every time.

- Only use 9V battery snap clips — no loose wires!
- No soldering or hot tools — ever!
- Ask your teacher before connecting the battery.

WHAT DID WE LEARN?

3 Big Ideas from Today!

Today we explored capacitors, batteries, and LEDs — and made a light blink without any chip!

- 🪣 Capacitor stores electricity like a water tank.
- ⚡ Battery is main power — capacitor fills and pours fast!
- 💡 LED blinks from the capacitor's quick pour of energy!

THANK YOU & KEEP EXPLORING!

Keep Tinkering, Asking & Discovering!

You made the LED blink without a chip — you're a Master Tinkerer!